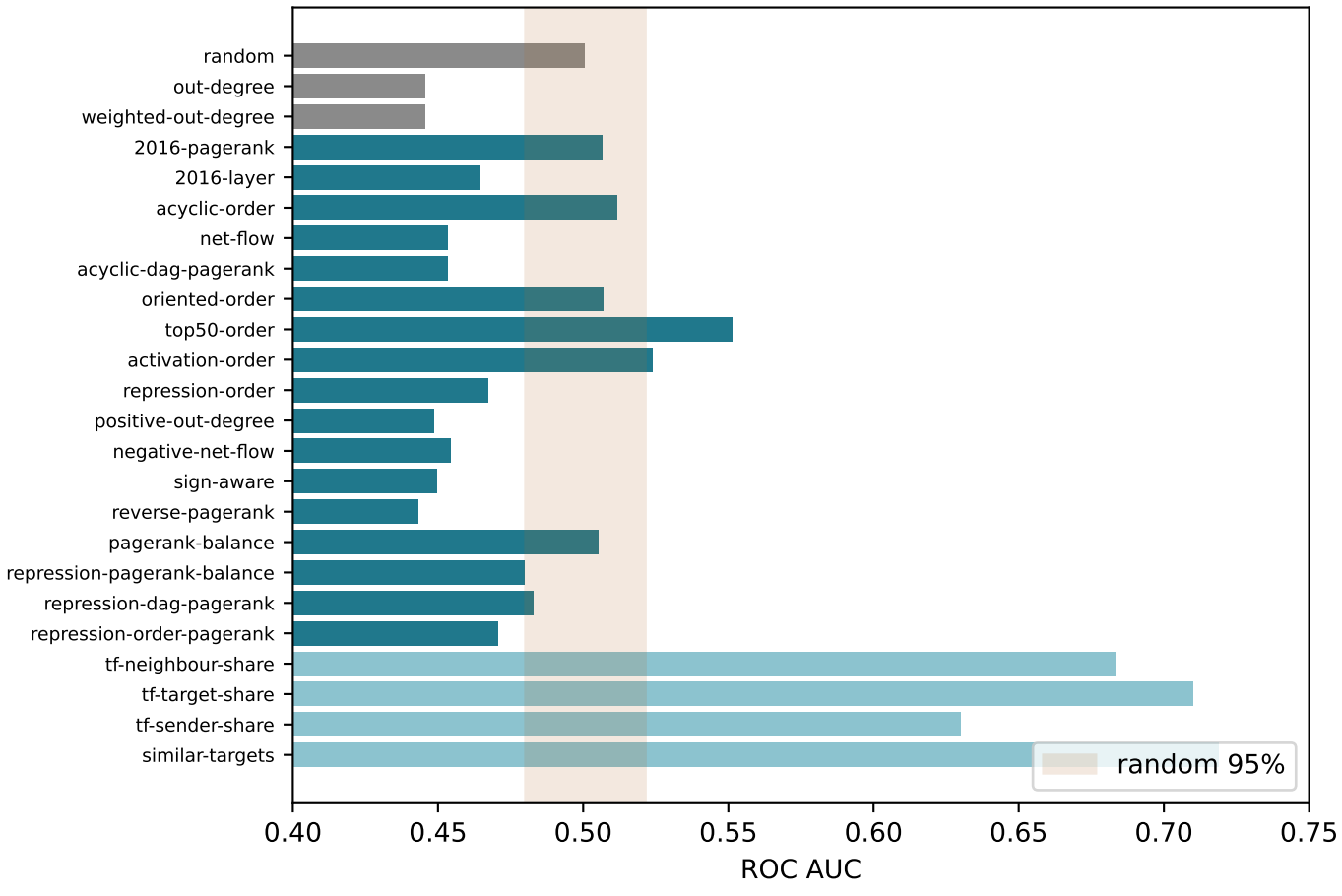
Regulator ranking against reference 'lambert-2018'

2026-11-10 · 8,945 genes ranked, 805 in lambert-2018-tfs.txt · top k = 100

Pipeline	Reference genes in top 100	AUC	p top-k	p AUC	AUC vs best baseline (95%)	p vs baseline
random	9.0 (4–15)	0.500 (0.480–0.521)				
out-degree	10	0.446	0.426	1.000	+0.000 (+0.000, +0.000)	
weighted-out-degree	11	0.446	0.301	1.000	-0.000 (-0.001, +0.001)	0.944
2016-pagerank	7	0.507	0.809	0.278	+0.061 (+0.031, +0.093)	< 0.001
2016-layer	6	0.464	0.893	1.000	+0.019 (+0.002, +0.036)	0.032
acyclic-order	10	0.512	0.426	0.148	+0.066 (+0.038, +0.093)	< 0.001
net-flow	11	0.453	0.301	1.000	+0.008 (-0.005, +0.020)	0.266
acyclic-dag-pagerank	11	0.453	0.301	1.000	+0.008 (+0.000, +0.016)	0.048
oriented-order	11	0.507	0.301	0.277	+0.061 (+0.031, +0.089)	< 0.001
top50-order	9	0.551	0.550	< 0.001	+0.106 (+0.073, +0.137)	< 0.001
activation-order	10	0.524	0.426	0.017	+0.078 (+0.050, +0.108)	< 0.001
repression-order	7	0.467	0.809	1.000	+0.022 (+0.001, +0.044)	0.036
positive-out-degree †	7	0.449	0.809	1.000	+0.003 (-0.003, +0.008)	0.344
negative-net-flow †	12	0.454	0.185	1.000	+0.009 (-0.007, +0.023)	0.276
sign-aware †	10	0.450	0.426	1.000	+0.004 (-0.006, +0.013)	0.450
reverse-pagerank †	8	0.443	0.691	1.000	-0.003 (-0.005, -0.000)	0.042
pagerank-balance †	10	0.505	0.426	0.320	+0.060 (+0.032, +0.087)	< 0.001
repression-pagerank-balance †	11	0.480	0.301	0.976	+0.034 (+0.010, +0.058)	0.008
repression-dag-pagerank †	11	0.483	0.301	0.952	+0.037 (+0.018, +0.057)	< 0.001
repression-order-pagerank †	8	0.471	0.691	0.999	+0.025 (+0.005, +0.046)	0.012
tf-neighbour-share ‡	61	0.683	< 0.001	< 0.001	+0.237 (+0.212, +0.261)	< 0.001
tf-target-share ‡	32	0.710	< 0.001	< 0.001	+0.264 (+0.237, +0.291)	< 0.001
tf-sender-share ‡	54	0.630	< 0.001	< 0.001	+0.184 (+0.158, +0.209)	< 0.001
similar-targets ‡	65	0.719	< 0.001	< 0.001	+0.273 (+0.242, +0.302)	< 0.001

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines. AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data. ‡: uses known TFs: scored from other genes' labels in this reference (lighter bars).



Edge checks against curated TF→target pairs

Recall ×: share of curated pairs present as edges, over chance.
Positive/negative ×: share of edges of that sign that are curated pairs, over chance. Sign: agreement with the curated sign (chance).

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Graph	Pairs	Recall (×)	Positive ×	Negative ×	Sign (chance)
raw graph	25,804	0.046 (1.12×)	3.99×	0.56×	0.58 (0.56)
resolve_reciprocals → drop_sinks → prune_triangles	25,705	0.016 (0.70×)	7.99×	0.64×	0.23 (0.23)
resolve_reciprocals → drop_sinks → prune_triangles → greedy_layering	6,926	0.047 (0.65×)	13.27×	0.64×	0.17 (0.18)
absolute	25,804	0.046 (1.12×)			
absolute → minimum_feedback_order	25,395	0.037 (0.91×)			
orient_reciprocals ratio=2.0 → absolute	25,653	0.035 (0.89×)			
orient_reciprocals ratio=2.0 → absolute → minimum_feedback_order	25,653	0.033 (0.84×)			
top_outgoing k=50 → absolute	25,804	0.013 (5.63×)			
top_outgoing k=50 → absolute → minimum_feedback_order	25,657	0.010 (5.19×)			
keep_sign sign=1 → absolute	25,241	0.028 (3.97×)			
keep_sign sign=1 → absolute → minimum_feedback_order	24,795	0.021 (3.90×)			
keep_sign sign=-1 → absolute	15,979	0.031 (0.46×)			
keep_sign sign=-1 → absolute → minimum_feedback_order	14,684	0.034 (0.46×)			
keep_sign sign=-1	15,979	0.031 (0.46×)		0.46×	0.19 (0.19)

Pipeline steps

out-degree: Most outgoing edges (raw graph)

score out_degree, descending

Graph: 3,770,673 edges (2,976,725 negative), largest strong component 8,900.

weighted-out-degree: Largest total outgoing weight (raw graph)

score weighted_out_degree, descending

Graph: 3,770,673 edges (2,976,725 negative), largest strong component 8,900.

2016-pagerank: 2016 cleaning, lowest PageRank

resolve_reciprocals() → drop_sinks() → prune_triangles() → score pagerank, ascending

Graph: 1,909,867 edges (1,889,772 negative), largest strong component 8,321.

2016-layer: 2016 cleaning, earliest greedy layer

resolve_reciprocals() → drop_sinks() → prune_triangles() → greedy_layering() → score level, ascending

Graph: 1,909,867 edges (1,889,772 negative), largest strong component 8,321; DAG keeps 1,784,787 edges.

acyclic-order: Earliest in acyclic order

absolute() → minimum_feedback_order() → score level, ascending

Graph: 3,770,673 edges (0 negative), largest strong component 8,900; DAG keeps 3,505,149 edges.

net-flow: Outgoing minus incoming weight

absolute() → score net_flow, descending

Graph: 3,770,673 edges (0 negative), largest strong component 8,900.

acyclic-dag-pagerank: Reverse PageRank on the acyclic DAG

absolute() → minimum_feedback_order() → score dag_reverse_pagerank, descending

Graph: 3,770,673 edges (0 negative), largest strong component 8,900; DAG keeps 3,505,149 edges.

oriented-order: Pairs with a 2x stronger side only, earliest in acyclic order

orient_reciprocals(ratio=2.0) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 3,407,077 edges (0 negative), largest strong component 8,004; DAG keeps 3,370,369 edges.

top50-order: Top 50 outgoing edges per gene, earliest in acyclic order

top_outgoing(k=50) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 276,017 edges (0 negative), largest strong component 6,373; DAG keeps 206,097 edges.

activation-order: Activating edges only, earliest in acyclic order

keep_sign(sign=1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 793,948 edges (0 negative), largest strong component 5,897; DAG keeps 573,044 edges.

repression-order: Repressing edges only, earliest in acyclic order

keep_sign(sign=-1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 2,976,725 edges (0 negative), largest strong component 1,881; DAG keeps 2,972,148 edges.

positive-out-degree: Most positive outgoing edges

score positive_out_degree, descending

Graph: 3,770,673 edges (2,976,725 negative), largest strong component 8,900.

negative-net-flow: Negative outgoing minus incoming weight

score negative_net_flow, descending

Graph: 3,770,673 edges (2,976,725 negative), largest strong component 8,900.

sign-aware: Rank sum: positive out-degree and negative net flow

score sign_aware, descending

Graph: 3,770,673 edges (2,976,725 negative), largest strong component 8,900.

reverse-pagerank: Highest reverse PageRank (raw graph)